PVP 23

Code: 23CS3302, 23IT3302, 23AM3302, 23DS3302

II B.Tech - I Semester – Regular Examinations - DECEMBER 2024

OBJECT ORIENTED PROGRAMMING THROUGH JAVA

(Common for CSE, IT, AIML, DS)

Duration: 3 hours Max. Marks: 70

Note: 1. This question paper contains two Parts A and B.
2. Part-A contains 10 short answer questions. Each Question carries 2 Marks.
3. Part-B contains 5 essay questions with an internal choice from each unit. Each Question carries 10 marks.
4. All parts of Question paper must be answered in one place.

BL – Blooms Level CO – Course Outcome

PART – A

		BL	CO
1.a)	Develop a program to implement Command Line Arguments in Java.	L3	CO1
1.b)	Explain the purpose of type casting in Java.	L2	CO1
1.c)	How do you access private members of a class? Explain.	L2	CO2
1.d)	Explain the methods used for searching with-in string in Java.	L2	CO2
1.e)	Explain how you can dynamically change the size of an array.	L2	CO2
1.f)	Define abstract class. Differentiate with concrete class.	L2	CO2
1.g)	Identify the differences between auto-boxing and auto-unboxing.	L2	CO3
1.h)	How does the Scanner class facilitate input	L2	CO3

	operations in Java? Explain.		
1.i)	Identify the main states in the Java thread life cycle.	L2	CO4
1.j)	What is the purpose of Collection Framework in Java?	L1	CO4

PART – B

			BL	CO	Max. Marks
UNIT-I					
2	a)	List and explain different data types supported by Java. Give suitable example program.	L2	CO1	5 M
	b)	Write a Java program to calculate the tax on a salary. The program should prompt the user to enter their annual salary. Use the following tax brackets: Sal< 2,50,000 rupees – No tax Sal > 2,50,001 and < 5,00,000 - 10% tax Sal> 5,00,000 – 5% tax Display the tax amount based on the entered salary.	L3	CO1	5 M
OR					
3	Explain the different types of operators in Java with examples.		L2	CO1	10 M
UNIT-II					
4	a)	Differentiate constructor overloading and method overloading. Give suitable examples.	L3	CO2	5 M

	b)	Explain how you can modify a string with example program.	L2	CO2	5 M
OR					
5	a)	Describe the process of declaring and initializing class and objects in Java with suitable example.	L2	CO2	5 M
	b)	Construct a Java program to differentiate passing arguments by Value and by Reference.	L3	CO2	5 M
UNIT-III					
6	a)	List and explain different operations that can be performed on Array elements.	L2	CO2	5 M
	b)	What is inheritance? Explain different types of inheritance techniques which are supported by Java.	L2	CO2	5 M
OR					
7	a)	Explain the following: i) final keyword in inheritance ii) Vector	L2	CO2	5 M
	b)	Discuss the concepts of default and static methods in Interface.	L2	CO2	5 M
UNIT-IV					
8	a)	How do different access control specifiers control access to class members across different packages? Explain.	L2	CO3	5 M
	b)	Differentiate between checked and unchecked exceptions. Give suitable examples.	L3	CO3	5 M

OR					
9	a)	Explain the usage of try, catch, throw, throws and finally keywords in exception handling. Give simple example.	L2	CO3	5 M
	b)	Illustrate the difference between byte streams and character streams in Java. Draw the stream hierarchies.	L3	CO3	5 M
UNIT-V					
10	a)	Explain how thread priority is set and used in Java with example program.	L4	CO4	5 M
	b)	Illustrate the use of HashSet class in collection framework with an example program.	L3	CO4	5 M
OR					
11	a)	Analyze different procedures for creating a thread in Java. Explain any one mechanism with example program.	L4	CO4	5 M
	b)	Define List and differentiate ArrayList, LinkedList.	L2	CO4	5 M

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II B.Tech - I Semester – Regular / Supplementary Examinations
NOVEMBER 2025

OBJECT ORIENTED PROGRAMMING THROUGH JAVA

(Common for CSE, IT, AIML, DS)

Duration: 3 hours Max. Marks: 70

Note: 1. This question paper contains two Parts A and B.
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4. All parts of Question paper must be answered in one place.

BL – Blooms Level CO – Course Outcome

PART – A

		BL	CO
1.a)	List any four Java buzzwords and explain their importance.	L2	CO1
1.b)	Differentiate between while and do-while loops.	L2	CO1
1.c)	Define the terms polymorphism and Encapsulation.	L2	CO1
1.d)	Explain any two methods to extract the characters from string.	L2	CO1
1.e)	What is an abstract class? How is it different from an interface?	L2	CO1
1.f)	How to declare and initialize a two-dimensional array?	L2	CO1
1.g)	Differentiate between throw and throws in exception handling.	L2	CO1

1.h)	Explain the use of the finally block in exception handling.	L2	CO1
1.i)	What is thread priority in Java? How is it set?	L2	CO1
1.j)	Distinguish between a thread and a process.	L2	CO1

PART – B

			BL	CO	Max. Marks
		UNIT-I			
2	a)	What are literals and symbolic constants in Java? Illustrate with examples.	L2	CO1	5 M
	b)	Write a Java program using nested if-else statements to determine the largest of three numbers.	L2	CO1	5 M
		OR			
3	a)	What are escape sequences in Java? Write a program that uses escape sequences to format output.	L2	CO1	3 M
	b)	Develop a program that calculates grades based on the marks for the following ranges: Range — Grade 90-100 — S 80-89 — A 70-79 — B 60-69 — C 50-59 — D 40-49 — E Less than 40 — Fail	L3	CO1	7 M

UNIT-II					
4	a)	Define classes and objects in Java. Write a program to create a class Student with attributes and methods to display details.	L3	CO2	5 M
	b)	Write a program to count the number of vowels, consonants, spaces, digits and special characters in a string.	L3	CO2	5 M
OR					
5	a)	What is a final method? How does it differ from a normal method?	L2	CO2	5 M
	b)	Develop a program to demonstrate how to access private members of a class.	L2	CO2	5 M
UNIT-III					
6	a)	Explain the use of the super keyword in inheritance.	L2	CO2	5 M
	b)	Differentiate between method overloading and method overriding with suitable examples.	L2	CO2	5 M
OR					
7	a)	Explain the concept of extending an interface with a suitable example.	L2	CO2	5 M
	b)	Write a Java program to accept a matrix of order NxN and interchange the diagonals.	L3	CO2	5 M
UNIT-IV					
8	a)	Explain the behavior of different access specifiers in packages with examples.	L2	CO3	5 M

	b)	A file contains 5 integers. Write a program to fetch the numbers from the file, and then calculate their sum, and append its sum to the file.	L3	CO3	5 M
OR					
9	a)	Write a program in Java to demonstrate the use of try, catch, and finally blocks.	L2	CO3	5 M
	b)	What are checked and unchecked exceptions? Explain with examples.	L2	CO3	5 M
UNIT-V					
10	a)	Write a Java program to create multiple threads by extending the Thread class.	L3	CO4	5 M
	b)	Explain thread life cycle with a neat sketch.	L2	CO4	5 M
OR					
11	a)	Explain how to insert and delete elements from a list collection with an example.	L2	CO4	5 M
	b)	Explain about hierarchy of collection interfaces.	L2	CO4	5 M